

Problem 8.17. Read the definition of *MIN-FORMULA* in Problem 7.46.

Part a. Show that *MIN-FORMULA* $\in$ PSPACE.

Proof.

□

Part b. Explain why this argument fails to show that *MIN-FORMULA* $\in$ coNP:

If $\phi \notin \text{MIN-FORMULA}$, then ϕ has a smaller equivalent formula. An NTM can verify that $\phi \in \overline{\text{MIN-FORMULA}}$ by guessing that formula.

Proof.

□